Pofatu Data Submission Guidelines

Thank you for contributing your data to the Pofatu Database.

These guidelines will help you fill in the Pofatu Data Submission Template.

The workbook contains 4 different spreadsheets to help catalog
(1) bibliographical sources,
(2) sample information,
(3) compositional data,
(4) analytical metadata.

Please do not edit spreadsheet 5 'Vocabularies'

For each sample, this information is stored and associated with different identifiers which are used as primary and foreign keys in our relational database:
3 **Source IDs** (data, artefact, site), 1 **Sample ID**, 1 **Artefact ID**, and 1 or more **Method IDs.**

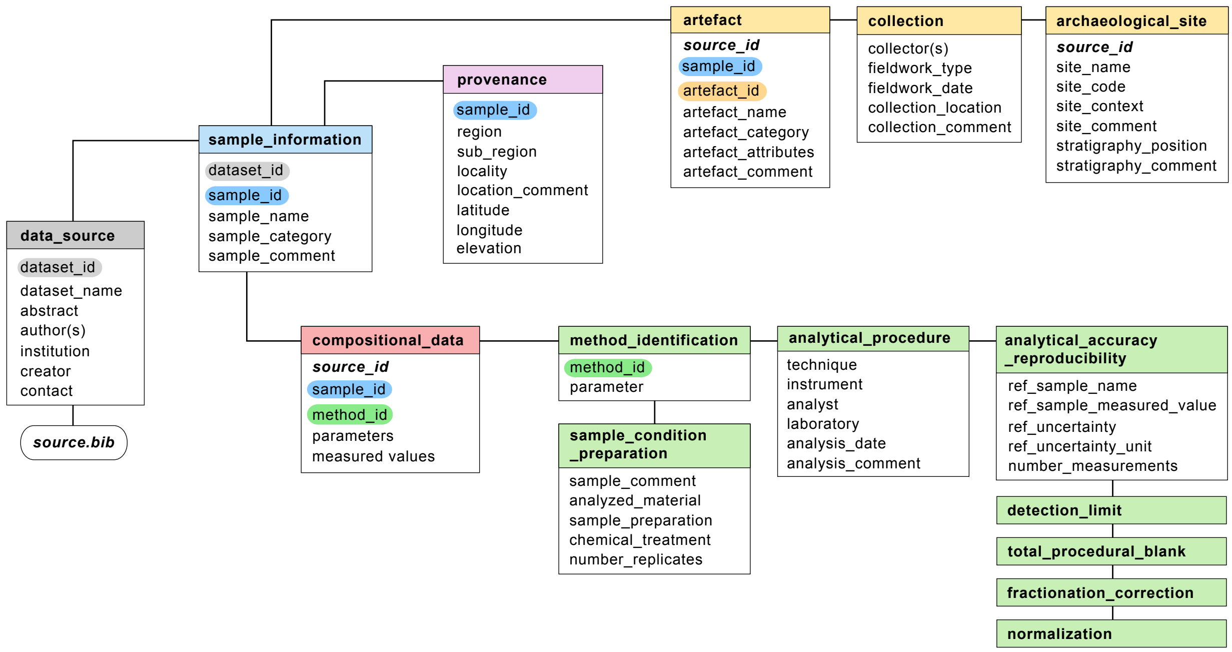
These identifiers are case and space sensitivity, and must respect the format indicated below.

These identifiers are case and space sensitive which will be edited during the data validation process. It is nonetheless necessary to provide consistent identifiers when contributing a new dataset. The authors of new datasets are asked to use the format below:

Dataset ID:	[first author's surname][publication year]
Source ID:	[first author's surname][publication year]
Sample ID:	[Source ID 1][_][Sample name]
Artefact ID:	[Artefact name][-][Site name]
Method ID:	[Source ID 1][_][A], [Source ID 1][_][B], etc.

Please let us know if your dataset contains duplicates (same sample name/ID measured more than once with the same analytical method).

Content and color code of the template:



Pofatu Data Submission Guidelines

1. Data source (* mandatory)

1.1. Dataset

Entry	Description	Required format
Dataset ID *	Synthetic citation of the dataset	[first author's surname][publication year]
Title *	Descriptive title of the dataset	<50 words
Abstract *	Brief description of dataset	<50 words
Authors(s) *	Author(s) of the dataset	Last name, First name
Institution of the author *	Institution(s) of the author(s)	Name of the institution; Country
Creator *	Person who fills out this template	Last name, First name
Contact info *	Contact of the creator of the template	Professional email address

1.2. Citation 1: Data

Entry	Description	Required format
Source ID 1 *	Short citation of the published geochemical data	[first author's surname][publication year]
Complete reference 1	Complete reference of the published geochemical data	Citation style: Nature
DOI 1	Digital Object Identifier of Complete reference 1	DOI if available

1.3. Citation 2: Artefact

Entry	Description	Required format
Source ID 2 *	Short citation of the published description of the parent artefact	[first author's surname][publication year]
Complete reference 2	Complete reference of the reference to the published description of the parent artefact	Citation style: Nature
DOI 2	Digital Object Identifier of Complete reference 2	DOI if available

1.4. Citation 3: Site

Entry	Description	Required format
Source ID 3 *	Short citation of the published description of the related archaeological site	[first author's surname][publication year]
Complete reference 3	Complete reference of the reference to the published description of the related archaeological site	Citation style: Nature
DOI 3	Digital Object Identifier of Complete reference 3	DOI if available

2. Sample information and archaeological provenance (* mandatory)

NB: In table 2 ('Sample Info & archaeological provenance'), several lines can be associated with the same Artefact ID if an artefact was sampled and analyzed more than once.

2.1.1. Sample identification

Entry	Description	Required format
Source ID 1 *	Short citation of the published geochemical data	Content must match citation code 1 in section 1.2
Sample name *	Original sample identification	Name given in the original publication of the dataset
Sample ID *	Unique ID of individual sample generated for Pofatu	[Citation code 1][_][Sample name]
Sample category *	Status of the analyzed sample	Drop-down list: Source, Artefact
Sample comment	Supplementary information about the status of the sample	eg. "Artefact used as a source"

2.1.2. Petrographic description

Entry	Description	Required format
Petrography	Petrographic description of the sample: texture, minerals, etc.	Short description (<50 words)
Comments	Supplementary information about the petrographic description of the sample	Short description (<50 words)

2.2. Archaeological provenance, description and location of parent artefacts and associated sites.

2.2.1. Provenance

Entry	Description	Required format
Region *	Name of the territory where the parent artefact was collected	Name of the archipelago, island group, etc.
Sub-region *	Name of the region where the parent artefact was collected	Name of the island, region, etc.
Locality *	Name of the locality where the parent artefact was collected	Site name, or village, valley, river, stream, etc.
Location comment	Supplementary information about the original location of the parent artefact	<20 words
Latitude *	Coordinates	Decimal degrees (negative for south of the equator)
Longitude *	Coordinates	Decimal degrees (negative for west of the prime meridian)
Elevation *	Altitude above the sea level	In meters

2.2.2. Artefact identification

Entry	Description	Required format
Source ID 2 *	Short citation of the published description of the parent artefact	Content must match Source ID 2 in section 1.3
Artefact ID *	Must be different from Sample ID. Make sure to use the same ID for the same artefacts	If no artefact identifier is provided (different from sample identifier), then create it : [Island][_][Sample name]
Artefact name *	Original artefact identification	Name given in the original publication of the parent artefact
Artefact category *	Typological category of the parent artefact	Drop-down list: Adze, Adze preform, Chisel, Cobble, Core, Flake, Geological, Grindstone, Hammerstone, NA
Artefact attributes	Fragmentation of the parent artefact	Drop-down list: Complete, Fragment (proximal, mesial, distal, or undetermined), Natural dyke, Natural boulder/cobble, Natural prism, NA
Artefact comments	Supplementary description of the parent artefact	< 50 words

2.2.3. Archaeological collection

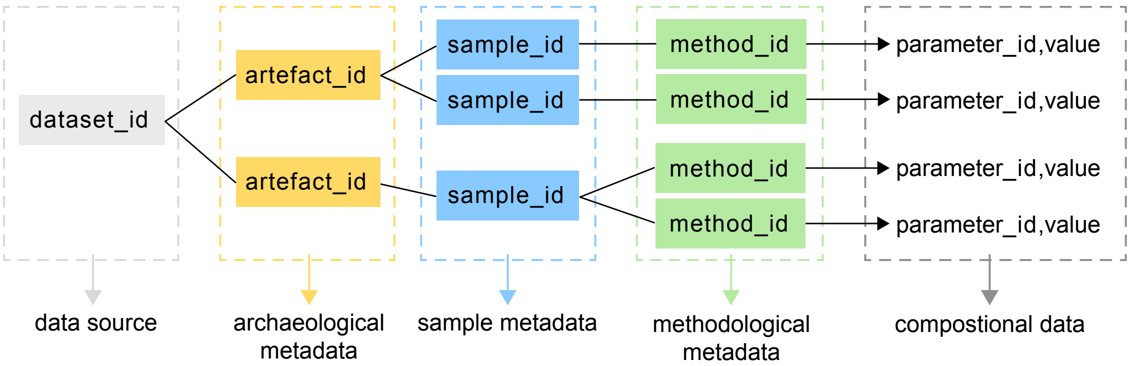
Entry	Description	Required format
Collector *	Principal investigator in charge of the fieldwork	Last name, First name
Fieldwork *	Type of field research	"Survey" or "Excavation"
Fieldwork date	Period of the field research	Year(s)
Collection storage location	Name of the institution managing the collection	<10 words
Collection comment	Supplementary information about the archaeological collection	<50 words

2.2.4. Site identification

Entry	Description	Required format
Source ID 3 *	Short citation of the related archaeological site	Content must match Source ID 3 in section 1.4
Site name	Original site identification 1	Name given in the original publication of the related archaeological site
Site code	Original site identification 2	Code given in the original publication of the related archaeological site
Site context *	Context of the related archaeological site	Drop-down list: Agricultural, Burial, Ceremonial, Domestic, Defensive, Midden, Natural, Quarry, Rockshelter, Workshop, NA
Site comments	Supplementary information about the archaeological site	eg. Precision about the archaeological context
Stratigraphic position *	Original stratigraphic context of the parent artefact	"Surface" or Stratigraphic unit, Layer, etc.
Stratigraphy comments	Supplementary information about the stratigraphic context	eg. Section, Square, etc.

3. Compositional data (* mandatory)

NB: In table 3 (‘Compositional data’), several lines can be associated with the same Sample ID if the same sample was analyzed more than once with different analytical techniques.



3.1. Sample & Method identification (content must match 2.1.1 and 4.1.1)

Entry	Description	Required format
Source ID 1 *	Short citation of the published geochemical data	Content must match Source ID 1 in section 1.2
Sample ID *	Unique ID of individual sample generated for Pofatu	Content must match Sample ID in section 2.1.1
Method code *	Unique identifier	Must match Method code in section 4.1

3.2. Measured values

Entry (fixed parameters)	Description	Required format
Major elements	Measured values of oxides (wt%)	Decimal separator is . (not ,)
Trace elements	Measured values of trace elements and REE (ppm)	Decimal separator is . (not ,) No separator of thousands
$^{143}Nd/^{144}Nd$ ratio	$^{143}Nd/^{144}Nd$ value $^{143}Nd/^{144}Nd$ Standard Deviation value $^{143}Nd/^{144}Nd$ SD sigma	Decimal separator is . (not ,) Decimal separator is . (not ,) 1σ or 2σ
$^{87}Sr/^{86}Sr$ ratio	$^{87}Sr/^{86}Sr$ value $^{87}Sr/^{86}Sr$ Standard Deviation value $^{87}Sr/^{86}Sr$ SD sigma	Decimal separator is . (not ,) Decimal separator is . (not ,) 1σ or 2σ
$^{206}Pb/^{204}Pb$ ratio	$^{206}Pb/^{204}Pb$ value $^{206}Pb/^{204}Pb$ Standard Deviation value $^{206}Pb/^{204}Pb$ SD sigma	Decimal separator is . (not ,) Decimal separator is . (not ,) 1σ or 2σ
$^{207}Pb/^{204}Pb$ ratio	$^{207}Pb/^{204}Pb$ value $^{207}Pb/^{204}Pb$ Standard Deviation value $^{207}Pb/^{204}Pb$ SD sigma	Decimal separator is . (not ,) Decimal separator is . (not ,) 1σ or 2σ
$^{208}Pb/^{204}Pb$ ratio	$^{208}Pb/^{204}Pb$ value $^{208}Pb/^{204}Pb$ Standard Deviation value $^{208}Pb/^{204}Pb$ SD sigma	Decimal separator is . (not ,) Decimal separator is . (not ,) 1σ or 2σ
$^{207}Pb/^{206}Pb$ ratio	$^{207}Pb/^{206}Pb$ value $^{207}Pb/^{206}Pb$ Standard Deviation value $^{207}Pb/^{206}Pb$ SD sigma	Decimal separator is . (not ,) Decimal separator is . (not ,) 1σ or 2σ
$^{208}Pb/^{206}Pb$ ratio	$^{208}Pb/^{206}Pb$ value $^{208}Pb/^{206}Pb$ Standard Deviation value $^{208}Pb/^{206}Pb$ SD sigma	Decimal separator is . (not ,) Decimal separator is . (not ,) 1σ or 2σ
Geochronology	Age (Ma) Age Standard deviation	Decimal separator is . (not ,) Decimal separator is . (not ,)

4. Primary analytical & method-specific metadata (* mandatory)

NB: In table 4 (‘Analytical metadata’), the same ‘Method ID’ will be associated with the same ‘Parameter’ if more than one ‘Reference sample’ is reported in section 4.1.2.

All the parameters documented in table 3 should be referenced and metadata should be provided in spreadsheet 4.

4.1. Method identification

Entry	Description	Required format
Method ID *	Identifier specific to the production of the compositional data	[Source ID 1][_][A], [Source ID 1][_][B], etc.
Parameter *	Geochemical parameter analyzed	Content must match the parameters in section 3.3

4.2. Individual sample condition and preparation

Entry	Description	Required format
Analyzed material 1 *	Description of the type of material analyzed	Drop-down list: Whole rock, Fused disk, etc.
Analyzed material 2 *	Description of the sample analyzed	Drop-down list: Core sample, Sample surface, etc.
Sample preparation	Primary transformation of the sample	e.g. "Crushed in agate mill, sieved at 500 microns"
Chemical treatment	e.g. whole rock samples were dissolved in 0.1M HF + 5.0M HNO3	e.g. "Whole rock samples were dissolved in 0.1M HF + 5.0M HNO3"
Number of replicates	If average of more than 1 value	Leave blank if only 1 value

4.3. Analytical procedure

Entry	Description	Required format
Technique *	Analytical technique	Drop-down list: XRF, EDXRF, WDXRF, ICP-AES, etc.
Instrument *	Measurement equipment	Brand and model of the machine
Laboratory *	Name of the institution	<10 words
Analyst *	Name(s) of the analyst(s)	Last name, First name
Analysis date *	Period of the analysis	Year(s)
Analysis comment	Supplementary information about the analytical procedure	eg. Reference to the method description in the literature

4.4. Analytical accuracy and reproducibility: Standards

Entry	Description	Required format
Reference sample name	Name of the international standard measured	Multiple entries for the same parameter must be created if more than 1 international standard is reported
Reference sample measured value	Measured value of the international standard. NB: Several international standards may be reported	Multiple entries for the same parameter must be created if more than 1 international standard is reported
Reference uncertainty	Standard deviation	Multiple entries for the same parameter must be created if more than 1 international standard is reported
Reference uncertainty unit		e.g. ppm, wt%
Number of measurements	If average of more than 1 value	Leave blank if only 1 value

4.5. Analytical accuracy and reproducibility: detection limit, blank, normalization

Entry	Description	Required format
Detection limit	Lower limit of detection	Decimal separator is . (not ,)
	Unit	e.g. ppm
Total procedural blank	Blank value	Decimal separator is . (not ,)
	Unit	e.g. ppm
Fractionation correction for isotopic analyses	Geochemical parameter analyzed	e.g. Nd146_Nd144
	Name of the sample measured	e.g. La Jolla
	Measured value of the sample	Decimal separator is . (not ,)
	Reference sample accepted value	Decimal separator is . (not ,)
Normalization	Reference sample name	e.g. JNdi-1
	Reference sample accepted value	Decimal separator is . (not ,)
	Citation	if known

For any questions or comments please contact aymeric.hermann@cnrs.fr

Vocabularies

This tab contains the suggested Vocabularies for data submission to Pofatu: Sample information, Artefact category and description, Site description, Parameter, Technique.

Contact us with questions or additions: aymeric.hermann@cnrs.fr

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SAMPLE CATEGORY	Sample Category description
SOURCE	Sample related to natural raw material, as a primary geological source location
ARTEFACT	Sample related to used material, transformed or not, that was removed and transported from its primary geological source location

ARTEFACT CATEGORY	Artefact Category description
ADZE	
ADZE PREFORM	
CHISEL	
COBBLE	Natural cobbles
CORE	
FLAKE	Undetermined flake
RETOUCHED FLAKE	Flake tool with retouched edge
RAW MATERIAL	Untransformed material transported from primary geological source to archaeological site
ARCHITECTURAL	Material transported from primary geological source and used as an element of architecture
GRINDSTONE	
PICK	
OVENSTONE	Stone (generally porphyric and vacuolar) used in earth oven
HAMMERSTONE	Stone used as hammer
NA	No information

ARTEFACT ATTRIBUTES	Artefact attributes description
COMPLETE	Complete artefact
FRAGMENT	Undetermined fragment of artefact
FRAGMENT (PROXIMAL)	Proximal end fragment (side of flake butt, or opposite side of point or cutting edge)
FRAGMENT (MESIAL)	Mesial part of artefact
FRAGMENT (DISTAL)	Distal end fragment (side of point or cutting edge)
NATURAL DYKE	Prismatic material extracted from dyke seam
NATURAL BOULDER/COBBLE	Redeposited material
NATURAL PRISM	Prismatic material (from dyke seam or other formations)
NA	No information

SITE CONTEXT	Site Context description

AGRICULTURAL
BURIAL
CEREMONIAL
DOMESTIC
DEFENSIVE
MIDDEN
NATURAL
QUARRY
ROCKSHELTER
WORKSHOP
NA

ANALYZED MATERIAL 1	Analyzed Material description
Whole rock	
Fused disk	
Volcanic glass	
Mineral	

ANALYZED MATERIAL 2	Analyzed Material description
Core sample	
Powder	
Probe sample	
Sample surface	
NA	

PARAMETER	Parameter description
AB	albite
AC	acmite
ACC	accessory minerals
ACETATE	acetate
ACT	actinolite
ACT-HORN	actinolitic hornblende
AEG	aegirine
Ag	silver
Age	sample age
AI	age item
Al	aluminum
Al2O3	aluminium oxide
ALK	alkalinity
ALKENONE	alkenone
ALLA	allanite
ALM	almandine
AL-SAP	aluminum saponite
ALT	alteration product
AMES	amesite

AMG	amygdule
AMPH	amphibole
AM-SI	amorphous silica
AN	anorthite
ANAL	analcite
ANCL	anorthoclase
AND	andalusite
ANDR	andradite
ANH	anhydrite
ANOR	anorthosite
APA	apatite
Ar	argon
Ar36	argon isotope ar36
Ar36_Ar38	ratio of argon isotopes ar36 to ar38
Ar36_Ar40	ratio of argon isotopes ar36 to ar40
Ar37_Ar40	ratio of ar37 to ar40
Ar38	argon isotope ar38
Ar38_Ar36	ratio of argon isotopes ar38 to ar36
Ar39	argon isotope ar39
Ar39_Ar40	ratio of argon isotopes ar39 to ar40
Ar40	argon isotope ar40
Ar40*	radiogenic ar40
Ar40_Ar36	ratio of argon isotopes ar40 to ar36
Ar40_Ar36(I)	initial ar40 to ar36 ratio
Ar40_Ar39	ratio of argon isotopes ar40 to ar39
Ar40_ATM	atmospheric argon isotope ar40
Ar40_He4	ratio of argon isotopes ar40 to helium isotope he4
Ar40_RG	radiogenic ar40
Ar40RG_Ar39K	ratio of radiogenic ar40 to ar39 produced from k
ARA	aragonite
ARMA	armalcolite
As	arsenic
Au	gold
AUG	augite
AWA	awaruite
B	boron
B_Ca	ratio of boron to calcium
B11_B10	boron isotope
Ba	barium
Ba_Ca	ratio of barium to calcium
BaO	barium oxide
BaSO4	barite
Be	beryllium
Be10	beryllium isotope be10
Be10_Be9	ratio of beryllium isotopes be10 to be9

Be10_Be9(T)	ratio of isotopes be10 to be 9, time corrected
Be9	beryllium isotope be9
Bi	bismuth
BIO	biotite
Br	bromium
BROMIDE	bromide
BST	bastite
BZT	bronzite
C	carbon
C(INORG)	inorganic carbon
C(ORG)	organic carbon
C(TOT)	total carbon
C_(non-carb)	non-carbonate carbon
C_N	ratio of organic carbon to total nitrogen
C14_AGE	c14_age
Ca	calcium
Ca_K	ratio of calcium to potassium
CaCO3	calcium carbonate
CALC	calcite
CALCITE	calcite
CALIBRATED C14_AGE	calibrated c14_age
CaO	calcium oxide
Ca-OL	calcium-olivine
CAP_DELTA_S33	deviation of delta_s33 from the terrestrial fractionation array, calculated as delta_s33 minus 1000 times (delta_s34 divided by 1000 then plus one) taken to the 0.515th power) minus 1 or cap_delta_33s=delta_s33-1000×[(delta_s34/1000+1)^0.515-1]
CARB	carbonate
CASU	calcium sulfate
CATS	calcium tschermak molecule
CCR	calcium chromium clinopyroxene
Cd	cadmium
Cd_Ca	ratio of cadmium to calcium
Ce	cerium
Ce136_Ce142	ratio of cerium isotopes ce136 to ce142
Ce138_Ce142	ratio of cerium isotopes ce138 to ce142
Ce140_Nd146	ratio of cerium to neodymium isotopes ce140 to nd146
Ce2O3	cerium oxide
CEL	celadonite
CEL-NON	celadonite-nontronite
CEL-NON-SAP	celadonite-nontranite-saponite
CELS	celsian
CH4	methane
CHL/SMEC	chlorite/smectite
CHLOR	chlorite

CHLORINITY	chlorinity
CHLORITE	chlorite
CHLOR-SAP	chlorite-saponite
CHRG_BAL	electrical charge balance, calculated as the sum of cations vs. anions
CHROM	chromite
CHRYS	chrysotile
Cl	chlorine
CLAY	clay
CLS	celestine
CLY	clay mineral
CMG	cummingtonite
Co	cobalt
CO1	carbon monoxide
CO2	carbon dioxide
CO2_umol	carbon dioxide measured in micromols
CoO	cobalt oxide
COR	corundum
CORD	cordierite
CORR	corrensite
CPX	clinopyroxene
CPY	chalcopyrite
Cr	chromium
Cr2O3	chromium oxide
CRICH	crichtonite
CRIS	cristobalite
Cs	cesium
Cs137_ACTIVITY	cesium isotope cs137 reported as activity
CSO	casio3, high pressure
CST	coesite
CTD	chloritoid
CTI	calcium titanium tschermak molecule
Cu	copper
Cu2O	copper(i) oxide
CUB	cubanite
CuO	copper oxide
DELTA_B11	equals: (ratio of boron isotopes b11/b10 in the sample divided by ratio of b11/b10 in reference material srm951) minus 1, multiplied by 1000
DELTA_C13	equals: (ratio of carbon isotopes c13/c12 in the sample divided by ratio of c13/c12 of the pee dee belemnite) minus 1, multiplied by 1000
DELTA_Ca44	equals :(ratio of 44ca to 40ca sample) divided by (ratio of 44ca std to 40ca std) minus 1, multplied by 1000
DELTA_Cl37	equals: (ratio of chlorine isotopes cl37/cl35 in the sample divided by ratio of cl37/cl35 of the standard mean ocean chlorine) minus 1, multiplied by 1000

DELTA_D	equals: (ratio of hydrogen isotopes h2/h1 in the sample divided by ratio of h2/h1 in reference material) minus 1, multiplied by 1000
DELTA_Fe56	equals: (ratio of iron isotopes fe56/fe54 in the sample divided by ratio of fe56/fe54 of irmm-014) minus 1, multiplied by 1000
DELTA_Fe57	equals: (ratio of iron isotopes fe57/fe54 in the sample divided by ratio of fe57/fe54 of irmm-014) minus 1, multiplied by 1000
DELTA_Li6	equals: (ratio of lithium isotopes li6/li7 in the sample divided by ratio of li6/li7 of the standard) minus 1, multiplied by 1000
DELTA_Li7	equals: (ratio of lithium isotopes li7/li6 in the sample divided by ratio of li7/li6 of the standard) minus 1, multiplied by 1000
DELTA_Mg25	equals: (ratio of magnesium isotopes mg25/mg24 in the sample divided by ratio of mg25/mg24 in the reference material) minus 1, multiplied by 1000
DELTA_Mg26	equals: (ratio of magnesium isotopes mg26/mg24 in the sample divided by ratio of mg26/mg24 in the reference material) minus 1, multiplied by 1000
DELTA_N15	equals: (ratio of nitrogen isotopes n15/n14 in the sample divided by ratio of n15/n14 in the reference material) minus 1, multiplied by 1000
DELTA_O18	equals: (ratio of oxygen isotopes o18/o16 in the sample divided by ratio of o18/o16 of the standard) minus 1, multiplied by 1000
DELTA_S33	equals: (ratio of sulfur isotopes s33/s32 in the sample divided by ratio of s33/s32 in the reference material) minus 1, multiplied by 1000
DELTA_S34	equals: (ratio of sulphur isotopes s34/s32 in the sample divided by ratio of s34/s32 in reference material) minus 1, multiplied by 1000
DELTA_S34SO4	sulfur isotope of the sulfate. equals: (ratio of sulphur isotopes s34/s32 in the sample divided by ratio of s34/s32 in reference material) minus 1, multiplied by 1000
DELTA_U234	equals: (ratio of uranium isotopes u234/u238 in the sample divided by ratio of u234/u238 in the secular equilibrium) minus 1, multiplied by 1000
DELTA_Zn66	equals :(ratio of 66zn to 64zn sample) divided by (ratio of 66zn std to 64zn std) minus 1, multplied by 1000
DI	diopside
DIA	diamond
DIC	dissolved inorganic carbon
DOL	dolomite
Dy	dysprosium
Dy2O3	dysprosium oxide
E_Cd	epsilon 114/110cadmium
E_Ce	epsilon ce = [(138ce/142ce)sample/(138ce/142ce)chur - 1]x10e4
E_Hf	epsilon hf = [(176hf/177hf)sample/(176hf/177hf)chur - 1]x10e4
E_Hf(T)	epsilon hf = [(176hf/177hf)sample/(176hf/177hf)chur - 1]x10e4, time corrected
E_Nd	epsilon nd = [(nd143/nd144)sample/(nd143/nd144)reference - 1]x10e4
E_Nd(JUV)	epsilon nd(juv) = [(nd143/nd144)sample/(nd143/nd144)juv - 1]x10e4, where juv is juvinas achondrite

E_Nd(T)	epsilon nd = [(nd143/nd144)sample/(nd143/nd144)reference - 1]x10e4, time corrected
E_Sr	epsilon sr= [(sr87/sr86)sample/(sr87/sr86)reference - 1]x10e4
E_Sr(T)	epsilon sr= [(sr87/sr86)sample/(sr87/sr86)reference - 1]x10e4, time corrected
E_Tl205	epsilon thallium isotope 205
EDE	edenite
ED-HORN	edenitic hornblende
EN	enstatite
EP	epidote
Er	erbium
Eu	europium
F	fluorine
FA	fayalite
Fe	iron
Fe2O3	ferric iron oxide (tri-valent iron)
Fe2O3T	total iron oxide content reported as ferric (tri-valent) iron
Fe3_SUMFe	ratio of fe3+ to total fe
Fe3O4	
Fe3P_FeT	feric iron total iron ratio
FeCO3	siderite
Fe-Fe	iron ferrite = fefe2o4
FeO	ferrous iron oxide (di-valent)
FE-ORE	iron ore
FeOT	total iron oxide content reported as ferrous (di-valent) iron
FE-OX	fe-oxide
FeS2	iron sulfide
FETI-OX	iron titanium oxide
FO	forsterite
FORMATE	formate
FPER	ferropericlase
FS	ferrosilite
FSP	feldspar
G_Os	gamma osmium
G_Os(T)	gamma osmium, time corrected
Ga	gallium
GAL	galaxite = mnal2o4
GAR	garnet
Gd	gadolinium
Gd2O3	gadolinium oxide
Ge	germanium
GKL	geikilite
GL	glass
GM	groundmass
GM-GL	glassy groundmass

GOLD	gold
GONN	gonnardite
GPH	graphite
GRAPH	graphite
GROS	grossular
H	hydrogen
H(TOT)	total hydrogen
H2	hydrogen
H2O	water
H2OM	crystal water (h2o-)
H2OP	crystal water (h2o+)
H2S	hydrogen sulfide
H2S(TOT)	total sulfide
H4SiO4	silicic acid
HALITE	halite
HAUS	hausmannite
HAUY	hauyne
HCO3	bicarbonate
HCY	hercynite
He	helium
He3	helium isotope he3
He3_He4	ratio of helium isotopes he3 to he4
He3_He4(R/Ra)	ratio of helium isotopes he3 to he4 expressed relative to the helium isotope ratio of the atmosphere
He4	helium isotope he4
He4(NCC)	helium isotope he4 measured by neutron coincidence counting
He4_Ar40	ratio of helium isotope he4 to argon isotope ar40
He4_He3	ratio of helium isotopes he4 to he3
He4_Ne20	ratio of helium isotope he4 to neon isotope ne20
He4_Ne21	ratio of helium isotope he4 to neon isotope ne21
HEDN	hedenbergite
HEM	hematite
HERC	hercynite
HEU	heulandite
Hf	hafnium
Hf176_Hf177	ratio of hafnium isotopes hf176 to hf177
Hf176_Hf177(I)	ratio of hafnium isotopes hf176 to hf177 at initial time
Hf176_Hf177(T)	ratio of hafnium isotopes hf176 to hf177, time corrected
Hf177_Hf178	ratio of hafnium isotopes hf177 to hf178
Hf178_Hf177	ratio of hafnium isotopes hf178 to hf177
Hf179_Hf177	ratio of hafnium isotopes hf179 to hf177
Hf180_Hf177	ratio of hafnium isotopes hf180 to hf177
HfO2	hafnium oxide
Hg	mercury
HGAR	hydrogarnet

Ho	holmium
HORN	hornblende
HSCHL	hydroschorlomite
HYGR	hydrogrossular
I	iodine
IDD	iddingsite
ILL	illite
ILM	ilmenite
In	indium
INT	interstitial
IO	included in olivine
Ir	iridium
JAC	jacobsite
JD	jadeite
K	potassium
K2O	potassium oxide
KAER	kaersutite
KAOLINITE	kaolinite
KELY	kelyphite
KF	potassium feldspar
Kr	krypton
Kr78	krypton isotope kr78
Kr78_Kr84	ratio of krypton isotopes kr78 to kr84
Kr80	krypton isotope kr80
Kr80_Kr84	ratio of krypton isotopes kr80 to kr84
Kr82	krypton isotope kr82
Kr82_Kr84	ratio of krypton isotopes kr82 to kr84
Kr83	krypton isotope kr83
Kr83_Kr84	ratio of krypton isotopes kr83 to kr84
Kr84	krypton isotope kr84
Kr86	krypton isotope kr86
Kr86_Kr84	ratio of krypton isotopes kr86 to kr84
Kr90	krypton isotope kr90
KS	kalsilite
KY	kyanite
La	lanthanum
La2O3	lanthanum oxide
LAU	laumonite
LAW	lawsonite
LEU	leucite
LHER	lherzolite
Li	lithium
Li7_Li6	ratio of litium isotopes li7 toli6
LOI	loss on ignition

Lu	lutetium
Lu176_Hf177	ratio of lutetium isotope lu176 to hafnium isotope hf177
Lu176_Hf177(T)	ratio of lutetium isotope lu176 to hafnium isotope hf177, time corrected
Lu176_Lu177	ratio of lutetium isotopes lu176 to lu177
LW	lawsonite
MAF	mafic minerals
MARC	marcasite
Mg	magnesium
Mg_Ca	ratio of magnesium to calcium
MgCO3	magnesium carbonate
MGCR	magnesiochromite
MGF	mg-ferrite
MGH	maghemite
MgO	magnesium oxide
MICA	mica
MIL	millerite
Mn	manganese
Mn_Ca	ratio of manganese to calcium
Mn3O4	manganese tetroxide
MnCO3	manganese carbonate
MNCR	manganese chromite
MnO	manganese oxide
Mn-OL	manganese olivine
MNZ	monazite
Mo	molybdenum
MONT	montmorillonite
MS	matnesite
MT	magnetite
MULL	mullite
MUS	muscovite
MYR	myrmekite
N	nitrogen
N(A)	percent nitrogen in the a defect site of diamonds
N(B)	percent nitrogen in the b defect site of diamonds
N(ORG)	organic nitrogen
N(TOT)	total nitrogen
N2	nitrogen
N2[nmol]	nitrogen measured in nanomols
N2_Ar36	ratio of nitrogen isotope n2 to argon isotope ar36
N2_He3	ratio of nitrogen isotope n2 to helium isotope he3
Na	sodium
Na2O	sodium oxide
NAC	clinopyroxene end-member nacr (si2o6)
NATR	natrolite
Nb	niobium

Nb2O3	niobium oxide
Nb2O5	niobium pentoxide
NCU	native copper
Nd	neodymium
Nd_Ca	ratio of neodymium to calcium
Nd142_Nd144	ratio of neodymium isotopes nd142 to nd144
Nd143_Nd144	ratio of neodymium isotopes nd143 to nd144
Nd143_Nd144(I)	ratio of neodymium isotopes nd143 to nd144 at initial age
Nd143_Nd144(T)	ratio of neodymium isotopes nd143 to nd144, time corrected
Nd144_Nd146	ratio of neodymium isotopes nd144 to nd146
Nd145_Nd144	ratio of neodymium isotopes nd145 to nd144
Nd146_Nd142	ratio of neodymium isotopes nd146 to nd142
Nd146_Nd144	ratio of neodymium isotopes nd146 to nd144
Nd146_Nd145	ratio of neodymium isotopes nd146 to nd145
Nd148_Nd144	ratio of neodymium isotopes nd148 to nd144
Nd148O_Nd144O	ratio of neodymium nd148 oxide to neodymium nd144 oxide
Nd150_Nd144	ratio of neodymium isotopes nd150 to nd144
Nd2O3	neodymium oxide
Ne	neon
Ne20	neon isotope ne20
Ne20_Ne22	ratio of neon isotope ne20 to ne22
Ne21	neon isotope ne21
Ne21_He4	ratio of neon isotope ne21 to helium isotope he4
Ne21_Ne20	ratio of neon isotope ne21 to ne20
Ne21_Ne22	ratio of neon isotope ne21 to ne22
Ne22	neon isotope ne22
Ne22_Ne20	ratio of neon isotope ne22 to ne20
Ne23	neon isotope ne23
NH3	ammonia
NH4	ammonium
Ni	nickel
NiO	nickel oxide
NN	unknown
NO2	nitrite
NO3	nitrate
O	oxygen
O2	dioxygen
OH	hydroxide
OL	olivine
OL(R)	ol(r)
OLIG	oligoclase
OLPS	olivine pseudomorph
OLXC	olivine xenocryst
OMPH	omphacite
OPAL	opal

OPAQ	opaque mineral
OPQ	opaque mineral
OPX	orthopyroxene
OR	orthoclase
ORE	ore
Os	osmium
Os(I)	common osmium at initial age
Os184_Os188	ratio of osmium isotopes os184 to os188
Os186_Os188	ratio of osmium isotopes os186 to os188
Os187_Os186	ratio of osmium isotopes os187 to os186
Os187_Os188	ratio of osmium isotopes os187 to os188
Os187_Os188(I)	ratio of osmium isotopes os187 to os188 at initial age
Os187_Os188(T)	ratio of osmium isotopes os187 to os188, time corrected
Os188	osmium isotope os188
Os188_Os192	ratio of osmium isotopes os188 to os192
Os189_Os188	ratio of osmium isotopes os189 to os188
Os190_Os188	ratio of osmium isotopes os190 to os188
Os192_Os188	ratio of osmium isotopes os192 to os188
OX	oxide
OX-CEL	oxyhydroxide-celadonite
OX-PHYLL	oxyhydroxide-phyllosilicate
P	phosphorus
P(ORG)	phosphorous, organic
P2O5	phosphorus oxide
Pa	protactinium
Pa231	protactinium isotope pa231
Pa231_ACTIVITY	protactinium isotope pa231, reported as activity
Pa231_Th230	ratio of protactinium 231 to thorium 230
Pa231_U235_ACTIVITY	ratio of protactinium isotope pa231 to uranium isotope u235, reported as activity
Pa231_XS	excess protactinium 231
PAL	palagonite
PARA	paragonite
PARG	pargasite
Pb	lead
Pb204	lead isotope pb204
Pb204_Pb206	ratio of lead isotopes pb204 to pb206
Pb206	lead isotope 206
Pb206_Pb204	ratio of lead isotopes pb206 to pb204
Pb206_Pb204(I)	ratio of lead isotopes pb206 to pb204 at initial age
Pb206_Pb204(T)	ratio of lead isotopes pb206 to pb204, time corrected
Pb206_Pb207	ratio of lead isotopes pb206 to pb207
Pb206_Pb208	ratio of lead isotopes pb206 to pb208
Pb206_U235	ratio of lead isotope pb206 to uranium isotope u235
Pb206_U238	ratio of lead isotope pb206 to uranium isotope u238

Pb207_Pb204	ratio of lead isotopes pb207 to pb204
Pb207_Pb204(I)	initial pb207_pb204 ratio (age-corrected)
Pb207_Pb204(T)	ratio of lead isotopes pb207 to pb204, time corrected
Pb207_Pb206	ratio of lead isotopes pb207 to pb206
Pb207_Pb206(I)	ratio of lead isotopes pb207 to pb206 at initial age
Pb207_U235	ratio of lead isotope pb207 to uranium isotope u235
Pb208_Pb204	ratio of lead isotopes pb208 to pb204
Pb208_Pb204(I)	ratio of lead isotopes pb208 to pb204 at initial age
Pb208_Pb204(T)	ratio of lead isotopes pb208 to pb204, time corrected
Pb208_Pb206	ratio of lead isotopes pb208 to pb206
Pb208_Pb206(I)	ratio of lead isotopes pb208 to pb206 at initial age
Pb208_Th232	ratio of lead isotope pb208 to thorium isotope th232
Pb210	lead isotope pb210
Pb210_ACTIVITY	lead isotope pb210, reported as activity
Pb210_Ra226	ratio of lead isotope pb210 to radium isotope ra226
Pb210_U238	ratio of lead isotope pb210 to uranium isotope u238
Pb210_XS	excess lead 210
Pd	palladium
PECT	pectolite
PENT	pentlandite
PEROV	perovskite
PGN	pigeonite
pH	pH
PHEN	phengite
PHIL	philippsite
PHLOG	phlogopite
PHYLL	phyllosilicate
PI	physical item
PLAG	plagioclase
PLEO	pleonaste
PL-XC	plagioclase xenocryst
Po210	polonium isotope Po210
Po210_ACTIVITY	polonium isotope po210, reported as activity
Po210_Pb210	ratio of polonium isotope 210 to lead isotope 210
PO4	phosphate
Pr	praseodymium
PREH	prehnite
PRESS	pressure
PSB	pseudobrookite
Pt	platinum
Pt190_Os188	ratio of platinum isotope pt190 to osmium isotope os188
PUMP	pumpellyite
PX	pyroxene (not classified)
PY	pyrite
PYPH	pyrophanite

PYR	pyrope
PYRH	pyrrhotite
PYRPH	pyrophyllite
QUARTZ	quartz
QZ	quartz
r2	the correlation coefficient used to calculate the best-fit line through ambient seawater and the concentration of a given chemical species measured in the samples, plotted against Mg
Ra	radium
Ra226	radium isotope ra226
Ra226_ACTIVITY	radium 226, reported as activity
Ra226_Th230	ratio of radium isotope 226 to thorium isotope 230
Ra226_Th230_ACTIVITY	ratio of radium isotope 226 to thorium isotope 230, reported as activity
Rb	rubidium
Rb87_Sr86	ratio of rubidium isotope rb87 to strontium isotope sr86
Re	rhenium
Re187_Os186	ratio of rhenium isotope re187 to osmium isotope os186
Re187_Os188	ratio of rhenium isotope re187 to osmium isotope os188
Rh	rhodium
RHO	rhonite
Ru	ruthenium
RUT	rutile
S	sulfur
S(TOT)	total sulfur
S_(SLFI)	sulfur present as sulfide
S_PY	sulfur present in pyrite form
SAL	salinity
SAP	saponite
SAP-FE-OX	saponite-iron-oxide
SAP-GA	saponite-glauconite
Sb	antimony
Sc	scandium
Sc2O3	scandium oxide
SCAP	scapolite
Se	selenium
SERP	serpentine
Si	silica
Si(AMORPH)	amorphous silica
SID	sideromelane
SIL	silicate
SILM	sillimanite
SiO2	silica oxide
SIOH	sio2, high pressure
SLFA	sulfate
SLFI	sulfide

Sm	samarium
Sm147_Nd143	ratio of samarium isotope sm147 to neodymium isotope nd143
Sm147_Nd144	ration of samarium isotope sm147 to neodymium isotope nd144
Sm147_Nd146	ratio of samarium to neodymium isotopes sm147 to nd146
Sm2O3	samarium oxide
SMEC	smectite
SMEC-CEL	smectite-celadonite
Sn	tin
SND	sanidine
SO2	sulfur dioxide
SO3	sulfur trioxide
SO4	sulfate
SODA	sodalite
SP	spinel
SPES	spessartine
SPH	sphene
SPHL	sphalerite
Sr	strontium
Sr_Ca	ratio of sr to ca
Sr84_Sr86	ratio of strontium isotopes sr84 to sr86
Sr84_Sr88	ratio of strontium isotopes sr84 to sr88
Sr86_Sr88	ratio of strontium isotopes sr86 to sr88
Sr87_Sr86	ratio of strontium isotopes sr87 to sr86
Sr87_Sr86(I)	ratio of strontium isotopes sr87 to sr86 at initial age
Sr87_Sr86(T)	ratio of strontium isotopes sr87 to sr86, time corrected
Sr87_Sr88	ratio of strontium isotopes sr87 to sr88
Sr88_Sr86	ratio of strontium isotopes sr88 to sr86
SrO	strontium oxide
SS	sheet silicate
SYMP	symplectite
SYMT	symthite
Ta	tantalum
Ta2O5	tantalum oxide
TACH	tachylite
TALC	talc
Tb	terbium
Te	tellurium
TEMP	temperature
Th	thorium
Th227_ACTIVITY	th227 reported as activity
Th228_Th232	ratio of thorium isotopes th228 to th232
Th230	thorium isotope th230
Th230_ACTIVITY	thorium isotope th230, reported as activity
Th230_Th232	ratio of thorium isotopes th230 to th232
Th230_Th232_ACTIVITY	ratio of thorium isotopes th230 to th232, reported as activity

Th230_U238	ratio of thorium isotope th230 to uranium isotope u238
Th230_U238_ACTIVITY	ratio of thorium isotope th230 to uranium isotope u238, reported as activity
Th230_XS	excess thorium 230
Th232	thorium 232
Th232_ACTIVITY	thorium isotope th232, reported as activity
Th232_Pb204	ratio of thorium isotope th232 to lead isotope pb204
Th232_Th230	ratio of isotopes th232 to th230
Th232_Th230 ACTIVITY	ratio of isotopes th232 to th230 measured as activity
Th232_U238	ratio of thorium isotope th232 to uranium isotope u238
Th234	thorium 234
Th234_XS	thorium 234 excess
Th238_Th232_ACTIVITY	ratio of thorium isotopes th238 to th232, reported as activity
ThO	thorium oxide
ThO2	thorium oxide
THOM	thomsonite
Ti	titanium
TIC	total inorganic carbon
TI-MT	ti-magnetite
TI-NB	titanoniobate
TiO2	titanium oxide
Tl	thallium
Tl205_Tl203	ratio of thallium isotopes tl205 to tl203
Tm	thulium
TOC	total organic carbon
TREM	tremolite
TREV	trevorite
TRI	tridymite
TRO	troilite
U	uranium
U234	uranium 234
U234_ACTIVITY	uranium isotope u234, reported as activity
U234_U238	ratio of uranium isotopes u234 to u238
U234_U238 ACTIVITY	ratio of isotopes u234 to u235 measured as activity
U234_U238 ACTIVITY(T)	uranium 234 to uranium 238 measured as activity at time t
U234_U238_ACTIVITY	ratio of uranium isotopes u234 to u238, reported as activity
U235_Pb204	ratio of uranium isotope u235 to lead isotope pb204
U235_U234	ratio of isotopes u235 to u234
U238	uranium 238
U238_ACTIVITY	uranium isotope u238, reported as activity
U238_Pb204	ratio of uranium isotope u238 to lead isotope pb204
U238_Pb206	ratio of uranium isotope u238 to lead isotope pb206
U238_Th230	ratio of uranium isotope u238 to thorium isotope th230
U238_Th232	ratio of uranium isotope u238 to thorium isotope th232

U238_Th232_ACTIVITY	ratio of uranium isotope u238 to thorium isotope th232, reported as activity
UK'37	alkenone uk'37
ULSP	ulvospinel
UO2	uranium oxide
UVR	uvarovite
V	vanadium
V2O3	vanadium sesquioxide
V2O5	vanadium pentoxide
VES	vesicles
VES-F	filled vesicle
W	tungsten
WO	wollastonite
WO3	tungsten trioxide
Xe	xenon
Xe124	xenon isotope xe124
Xe124_Xe130	ratio of xenon isotopes xe124 to xe130
Xe124_Xe132	ratio of xenon isotopes xe124 to xe132
Xe126	xenon isotope xe126
Xe126_Xe130	ratio of xenon isotopes xe126 to xe130
Xe126_Xe132	ratio of xenon isotopes xe126 to xe132
Xe128	xenon isotope xe128
Xe128_Xe130	ratio of xenon isotopes xe128 to xe130
Xe128_Xe132	ratio of xenon isotopes xe128 to xe132
Xe129	xenon isotope xe129
Xe129_Xe130	ratio of xenon isotopes xe129 to xe130
Xe129_Xe132	ratio of xenon isotopes xe129 to xe132
Xe129_Xe136	ratio of xenon isotopes xe129 to xe136
Xe130	xenon isotope xe130
Xe130_Xe132	ratio of xenon isotopes xe130 to xe132
Xe131	xenon isotope xe131
Xe131_Xe130	ratio of xenon isotopes xe131 to xe130
Xe131_Xe132	ratio of xenon isotopes xe131 to xe132
Xe132	xenon isotope xe132
Xe132_Xe130	ratio of xenon isotopes xe132 to xe130
Xe134	xenon isotope xe134
Xe134_Xe130	ratio of xenon isotopes xe134 to xe130
Xe134_Xe132	ratio of xenon isotopes xe134 to xe132
Xe136	xenon isotope xe136
Xe136_He4	ratio of xenon isotope xe136 to helium isotope he4
Xe136_Xe130	ratio of xenon isotopes xe136 to xe130
Xe136_Xe132	ratio of xenon isotopes xe136 to xe132
Y	yttrium
Y2O3	yttrium oxide
Yb	ytterbium

Yb176_Hf177	ratio of ytterbium isotope yb176 to hafnium isotope hf177
YMG	yimengite
ZEOL	zeolite
ZIRC	zircon
Zn	zinc
ZnO	zinc oxide
ZO	zoisite
ZOI	zoisite
Zr	zirconium
Zr2O3	zirconium pentoxide
ZrO2	zirconium oxide

TECHNIQUE	Technique description
14C	CARBON 14 AGE DETERMINATION
AA	ACTIVATION ANALYSIS
AAS	ATOMIC ABSORPTION
AGE:ASSIG	Assigned Sample Age
AGE:GEO	Geological Sample Age
ALPHA	ALPHA COUNTING
ALPHA-ID	ALPHA COUNTING ISOTOPE DILUTION
AMS	ACCELERATOR MASS SPECTROMETRY
ANC	ANION CHROMATOGRAPHY
AR_AR	AR40_AR39 AGE DETERMINATION
ARC	ARC SPECTROMETRY
AUT	AUTO ANALYZER
AUTLCH	AUTOMATED LEACHING
BOMB	CARBONATE BOMB
CALC	CALCULATED
CCP	CALCIUM CARBONATE PRESERVATION
CHN	CARBON HYDROGEN NITROGEN ANALYSIS
CHN-G	CHN GAS CHROMATOGRAPHY
CHS	CARBON HYDROGEN SULFUR ELEMENTAL ANALYZER
CMBN	COMBUSTION
CNS	CARBON NITROGEN SULFUR ANALYZER
COL	COLORIMETRIC ANALYSIS
COUL	COULOMETRICAL ANALYSIS
CSA	CARBON SULFUR ANALYSIS
DCP	DIRECT CURRENT PLASMA (FURTHER DETAIL NOT PROVIDED)
DCP-AES	DIRECT CURRENT PLASMA ATOMIC EMISSION SPECTROMETRY
DROES	DIRECT READING OPTICAL EMISSIONS SPECTROSCOPY
EA	ELEMENTAL ANALYSIS
EA-CF-IRMS	ELEMENTAL ANALYZER-CONTINUOUS FLOW ISOTOPE RATIO MASS SPECTROMETRY
EMP	ELECTRON MICROPROBE
ENAA	EPITHERMAL NEUTRON ACTIVATION ANALYSIS

ES	EMISSION SPECTROMETRY
FIS	FISSION TRACK
FL-ES	FLAME EMISSION SPECTROSCOPY
FP	FILM POLAROGRAPHY
FPHOT	FLAME PHOTOMETRY
FTIR	FOURIER TRANSFORM INFRARED SPECTROMETRY
GAMMA	GAMMA COUNTING
GC	GAS CHROMATOGRAPHY
GC-MS	GAS CHROMATOGRAPHY- MASS SPECTROMETRY
GF-AAS	GRAPHITE FURNACE ATOMIC ABSORPTION SPECTROMETRY
GIO	GRADIENT ION-CHROMATOGRAPHY
GL-EL	GLASS ELECTRODE
GRAV	GRAVIMETRY
GS	GASOMETRIC TECHNIQUE
HPLC	HIGH-PERFORMANCE LIQUID CHROMATOGRAPHY
HR-ICP-MS	HIGH-RESOLUTION INDUCTIVELY COUPLED PLASMA MASS SPECTROMETRY
IA	IMAGE ANALYSIS
ICP	INDUCTIVELY COUPLED PLASMA (FURTHER DETAIL NOT PROVIDED)
ICP-AES	INDUCTIVELY COUPLED PLASMA ATOMIC EMISSION SPECTROSCOPY
ICP-ES	INDUCTIVELY COUPLED PLASMA SOURCE EMISSION SPECTROMETRY
ICP-LA	INDUCTIVELY COUPLED PLASMA LASER ABLATION (FURTHER DETAIL NOT PROVIDED)
ICP-LAM	INDUCTIVELY COUPLED PLASMA MASS SPECTROMETRY LASER ABLATION MICROPROBE
ICP-MS	INDUCTIVELY COUPLED PLASMA MASS SPECTROMETRY
ICP-MS-HR	INDUCTIVELY COUPLED PLASMA MASS SPECTROMETRY HIGH-RESOLUTION
ICP-MS-ID	INDUCTIVELY COUPLED PLASMA MASS SPECTROMETRY ISOTOPE DILUTION
ICP-MS-LA	INDUCTIVELY COUPLED PLASMA MASS SPECTROMETRY LASER ABLATION
ICP-MS-LA-MC	INDUCTIVELY COUPLED PLASMA MASS SPECTROMETRY LASER ABLATION MULTICOLLECTOR
ICP-MS-MC	INDUCTIVELY COUPLED PLASMA MASS SPECTROMETRY MULTI-COLLECTOR
ICP-OES	OPTICAL EMISSION SPECTROSPCOPY
ICS	IRIDIUM COINCIDENCE SPECTROMETRY
IGN	IGNITION
IMP	ION MICROPROBE
INAA	INSTRUMENTAL NEUTRON ACTIVATION ANALYSIS
INC	ION CHROMATOGRAPHY
IR-SP	INFRA-RED SPECTROSCOPY

ISE	ION SENSITIVE ELECTRODE
IVA	INVERSION VOLT-AMPERMETRY
K-AR	K-AR AGE DETERMINATION
LA-ICP	INDUCTIVELY COUPLED PLASMA LASER ABLATION (FURTHER DETAIL NOT PROVIDED)
LA-ICP-MS	LASER ABLATION INDUCTIVELY COUPLED PLASMA MASS SPECTROMETRY
LF	LASER FLUORINATION
LU-HF	LUTETIUM-HAFNIUM AGE DETERMINATION
MANO	MANOMETRY
MBS	MOLYBDATE-BLUE SPECTROMETRY
MC-ICP-MS	LASER ABLATION MULTICOLLECTOR INDUCTIVELY COUPLED PLASMA MASS SPECTROMETRY
MOSS	MOSSBAUER SPECTRA
MS	MASS SPECTROMETRY
MS:ID	ISOTOPE DILUTION MASS SPECTROMETRY
MS:SIMS	SECONDARY IONIZATION MASS SPECTROMETRY
MS:SS-ID	SPARK SOURCE MASS SPECTROMETRY - ISOTOPE DILUTION
MS:SSMS	SPARK SOURCE MASS SPECTROMETRY
MS:TIMS	THERMAL IONIZATION MASS SPECTROMETRY
MS:TIMS-ID	TIMS ISOTOPE DILUTION
MS:TIMS-ID-NEG	NEGATIVE THERMAL IONIZATION MASS SPECTROMETRY ISOTOPE DILUTION
MS:TIMS-NEG	NEGATIVE THERMAL IONIZATION MASS SPECTROMETRY
MS-ID	MASS SPECTROMETRY ISOTOPE DILUTION
NAA	NEUTRON ACTIVATION ANALYSIS
NAA-PG	PROMPT-GAMMA NEUTRON ACTIVATION ANALYSIS
NCC	NEUTRON COINCIDENCE COUNTING
NN	UNKNOWN
NSFA-ID	NICKEL SULFIDE FIRE ASSAY ISOTOPE DILUTION
NTIMS	NEGATIVE THERMAL IONIZATION MASS SPECTROMETRY
NTIMS-ID	NEGATIVE THERMAL IONIZATION MASS SPECTROMETRY ISOTOPE DILUTION
OES	OPTICAL EMISSION SPECTROMETRY
OPS	OPTICAL SPECTROSCOPY
PB-PB	PB-PB DATING
PC	POINT-COUNTING
PEN	PENFIELD METHOD
PGNAA	PROMPT-GAMMA NEUTRON ACTIVATION ANALYSIS
PIXE-PIGME	PROTON INDUCED X-RAYS AND GAMMA RAY EMISSION ANALYSIS
PMP	PROTON MICROPROBE
POT	POTENTIOMETRY
PYHY	PYROHYDROLYSIS

RA-TH	RA226_TH230 AGE DETERMINATION
RB-SR	RB-SR AGE DETERMINATION
REF	REFRACTOMETER
RE-OS	RHENIUM-OSMIUM AGE DETERMINATION
REP	ROCK EVAL PYROLYSIS
RNAA	RADIOANALYTICAL NEUTRON ACTIVATION
RR	RAPID ROCK
RSA	RADIO-ISOTOPIC SAMPLE AGE
SEM	SCANNING ELECTRON MICROSCOPE
SEM-EDS	SEM-ENERGY DISPERSIVE XRAYS
SIMS	SECONDARY ION MASS SPECTROMETRY
SMA	SOLID'S MOISTURE ANALYSIS
SM-ND	SAMARIUM-NEODYMIUM AGE DETERMINATION
SPEC	SPECTROGRAPHIC ANALYSIS
SP-PH	SPECTROPHOTOMETRY
SS-ID	SPARK SOURCE MASS SPECTROMETRY - ISOTOPE DILUTION
SSMS	SPARK SOURCE MASS SPECTROMETRY
TH_PB	THORIUM-LEAD AGE DETERMINATION
TIMS	Thermal Ionization Mass Spectrometer
TIMS-ID	THERMAL IONIZATION MASS SPECTROMETRY ISOTOPE DILUTION
TITR	TITRATION
U_PA	U235_PA231 AGE DETERMINATION
U_PB	URANIUM LEAD AGE DETERMINATION
U_TH	U238_TH230 AGE DETERMINATION
U_TH_HE	U-TH-HE AGE DETERMINATION
UNKNOWN	UNKNOWN
U-PB	URANIUM-LEAD AGE DETERMINATION
UV-ES	ULTRAVIOLET EMISSION SPECTROGRAPHY
VAC-F	VACUUM FUSION
VOL	VOLUMETRIC ANALYSIS
WET	WET CHEMISTRY
XANES	X-RAY ABSORPTION NEAR-EDGE STRUCTURE SPECTROSCOPY
XRD	X-RAY DISPERSIVE SPECTROMETRY
XRF	X-RAY FLUORESCENCE
XRF-EDS	ENERGY-DISPERSIVE X-RAY FLUORESCENCE
XRF-SCAN	X-RAY FLUORESCENCE SCANNING
ED-XRF	ENERGY DISPERSIVE XRF
WD-XRF	WAVELENGH DISPERSIVE XRF

Standard deviation
1σ
2σ